FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

NAME : KOLLURI SRUTHI PROBLEM STATEMENT NO: 42

SRN : PES1UG24CS230 SECTION : D

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ping microservice health endpoints at **30-second intervals** and compute rolling **24-hour service availability percentages**.	High	**Pass:** Health probes are executed every 30 seconds and the system calculates a 24-hour availability percentage. **Fail:** Probes are not performed at the specified interval or availability is not calculated.	Ensures continuous monitoring and availability tracking.
FR-002	Functional	The system shall maintain and display the **dependency relationships between microservices**.	High	**Pass:** When a service is selected, its dependent and dependency services are displayed correctly. **Fail:** A configured dependency is missing or incorrectly displayed.	Helps architects and DevOps engineers understand service architecture.
FR-003	Functional	The system shall aggregate and provide access to **API documentation** associated with each registered microservice.	High	**Pass:** Selecting a registered service displays its available API documentation. **Fail:** The documentation cannot be retrieved for a service with valid documentation.	Provides developers and administrators with centralized API information.

FR-004	Functional	The system shall record service downtime when **three consecutive health probes fail** and dispatch a downtime alert.	High	**Pass:** After three consecutive failed probes, downtime is recorded and an alert is dispatched. **Fail:** The service remains healthy or no alert is generated after three consecutive failures.	Enables rapid detection and notification of service outages.
FR-005	Functional	The system shall allow authorized users to **view the health status and availability percentage** of registered microservices.	Medium	**Pass:** An authorized user can select a service and view its current health status and 24-hour availability percentage. **Fail:** The status or availability information is unavailable or incorrect.	Gives stakeholders a centralized view of service health.
NFR-001	Nonfunctional	The service catalog dependency graph viewer shall render interactive node architectures containing up to **200 services** smoothly.	High	**Pass:** Benchmarking tests confirm that dependency graphs containing up to 200 services meet the defined performance and security standards under	Ensures the dependency graph remains usable for large service architectures.

				simulated peak load. **Fail:** The graph becomes unusable or violates the tested performance/security targets.	
NFR-002	Nonfunctional	The system shall restrict access to the microservice catalog and health information to **authenticated and authorized users**.	High	**Pass:** An unauthenticated or unauthorized user is denied access to protected portal functions. **Fail:** A user without required authorization can access protected information.	Protects internal service information and operational data.

USE CASE FLOW:

Preconditions

1. The microservice is registered in the system.
2. The microservice has a valid health endpoint.
3. The health monitoring service is active.
4. The DevOps Engineer has access to the portal.

Postconditions

Success:

- The health status of the microservice is updated.
- The health-check result is recorded.
- The service availability is calculated and updated.

Failure:

- The failed health check is recorded.
- If three consecutive health checks fail, a downtime alert is received.

Main Success Scenario

1. The DevOps Engineer opens the **Monitor Microservice Health** function.
2. The system selects the registered microservice to be monitored.
3. The system invokes **Check Health Endpoint (UC-08)**.
4. The system sends a health-check request to the microservice.
5. The microservice returns a successful health response.
6. The system records the successful health-check result.
7. The system invokes **Calculate Availability (UC-09)**.
8. The system calculates the microservice's rolling 24-hour availability.
9. The system updates and displays the current health status and availability.
10. The monitoring process continues with the next scheduled health check.